

PORTFOLIO PROJECT REPORT



AGRICULTURE PRODUCTION INSIGHTS & ANALYTICS

A Power BI Dashboard & Data Analysis Report

Prepared & Created by

NEHA BHATT

Business Analyst | Data Analytics Portfolio

Nadiad, Gujarat, India

August 2026

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1. Executive Summary

This report presents an advanced Power BI analysis of agriculture production data spanning four regions, five major crops, and a four-year period (2020–2023). The project was undertaken to convert raw agricultural records into a decision-ready dashboard that highlights production trends, yield efficiency, and the environmental and input factors that influence farm output.

The underlying dataset contains 10,000 records covering area harvested, yield, total production, rainfall, temperature, soil pH, fertilizer usage, and pest incidents. Analysis of this data shows that total production is broadly balanced across regions and crops, with the North region and Corn crop leading marginally, while yield per hectare shows only a weak relationship with rainfall, temperature, fertilizer use, or pest incidents — pointing to other agronomic and management factors as the real drivers of yield.

Total Records	Total Production	Avg. Yield/ha	Regions × Crops
10,000	19.25 Million Tons	3.48 Tons/ha	4 × 5

2. Project Objective

The objective of this project was to design an advanced, interactive Power BI dashboard that enables stakeholders to:

- Track total agricultural production across regions, crops, and years.
- Compare yield efficiency (tons per hectare) between crops and regions.
- Understand how environmental factors — rainfall, temperature, and soil pH — relate to yield outcomes.
- Monitor input usage (fertilizer) and risk factors (pest incidents) alongside output.
- Support data-driven planning for resource allocation and crop selection.

3. Dataset Overview

The analysis is based on the "Agriculture Production Insights" dataset, comprising 10,000 records across 11 fields. Each record represents the production performance of a specific crop, in a specific region, for a given year.

Field	Description	Type
Region	East, West, North, South	Categorical
Year	2020 – 2023	Numeric (Year)
Crop	Barley, Corn, Soybean, Rice, Wheat	Categorical
Area Harvested (ha)	Land area used for cultivation	Numeric
Yield per ha (tons)	Output efficiency per hectare	Numeric
Total Production (tons)	Total crop output	Numeric
Avg. Rainfall (mm)	Seasonal rainfall received	Numeric
Avg. Temperature (°C)	Seasonal average temperature	Numeric
Soil pH	Soil acidity/alkalinity level	Numeric
Fertilizer Used (kg)	Fertilizer input quantity	Numeric
Pest Incidents	Count of recorded pest events	Numeric

4. Tools & Methodology

Microsoft Power BI was used as the primary tool for data modelling, DAX-based calculations, and interactive dashboard design. The workflow followed a structured, phase-wise approach:

- Data import and profiling of the CSV source in Power Query.
- Data cleaning: type validation, handling of duplicates, and column standardisation.
- Data modelling and creation of calculated columns / measures using DAX.
- Design of interactive visuals: bar charts, trend lines, KPI cards, and slicers for Region, Crop, and Year.
- Formatting and UX polish for an executive-ready dashboard layout.

Supplementary statistical analysis (aggregation and correlation) was performed on the source dataset to validate and support the dashboard's visual findings, and is presented in the section below.

5. Key Findings & Insights

5.1 Production by Region

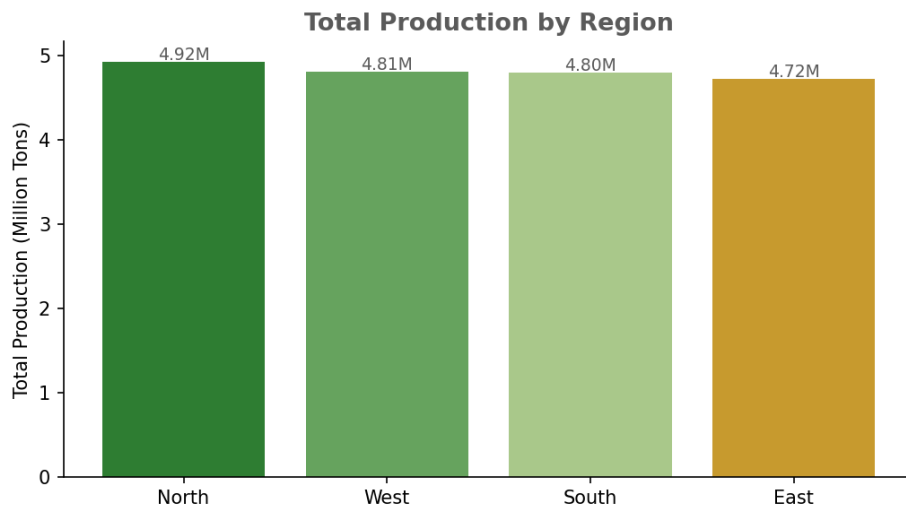


Figure 1: Total Production by Region (Million Tons)

Production is closely matched across all four regions. The North region leads with 4.92 million tons, followed by West (4.81M) and South (4.80M), with East slightly behind at 4.72 million tons. The gap between the highest and lowest region is under 5%, indicating a well-distributed national production base rather than dependence on any single region.

5.2 Production by Crop

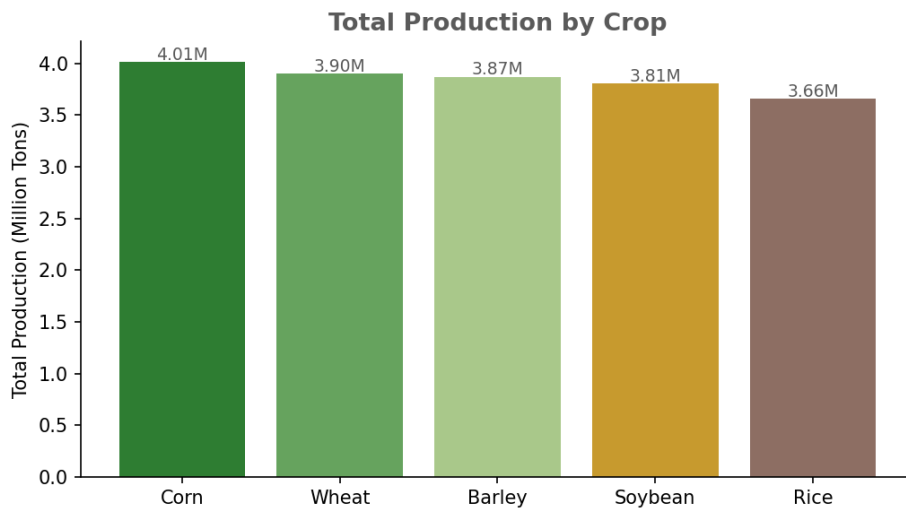


Figure 2: Total Production by Crop (Million Tons)

Corn contributes the highest total production at 4.01 million tons, followed by Wheat (3.90M) and Barley (3.87M). Soybean and Rice follow closely behind at 3.81 million and 3.66 million tons respectively. All five crops fall within a similar range, showing a diversified and balanced cropping pattern across the dataset.

5.3 Yearly Production Trend

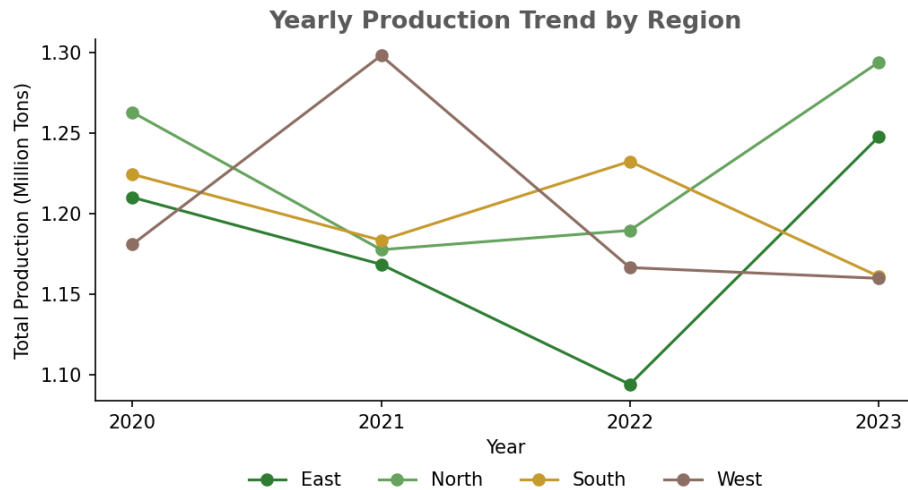


Figure 3: Yearly Production Trend by Region (Million Tons)

Total production across all regions has remained stable year-on-year, ranging narrowly between 4.68 and 4.88 million tons from 2020 to 2023. No region shows a sustained upward or downward trend, suggesting production levels are governed more by yearly variability in local conditions than by any consistent multi-year growth or decline pattern.

5.4 Yield Efficiency by Crop

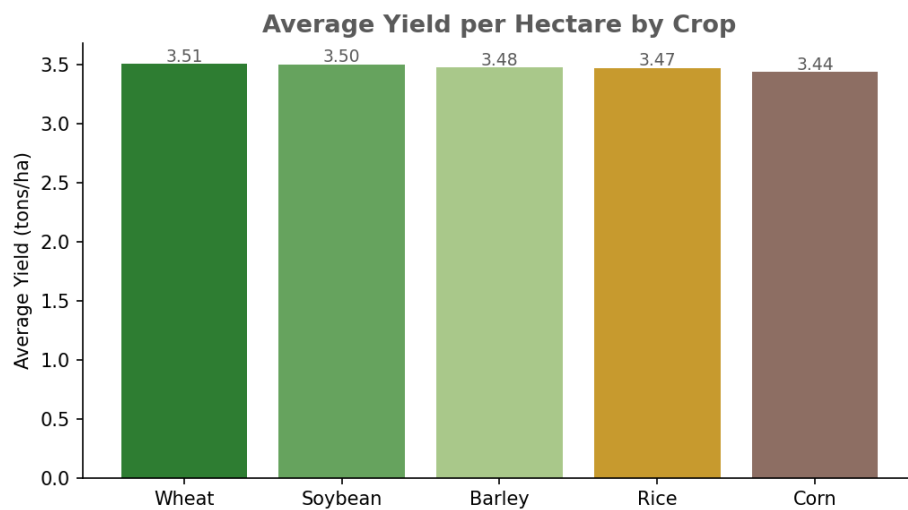


Figure 4: Average Yield per Hectare by Crop

Wheat records the highest average yield efficiency at 3.51 tons/ha, marginally ahead of Soybean (3.50), Barley (3.48), and Rice (3.47). Corn, despite having the highest total production, has the lowest average yield per hectare (3.44 tons/ha) — implying its output volume is driven more by larger area under cultivation than by superior per-hectare efficiency.

5.5 What Drives Yield?

A correlation analysis was run between Yield per Hectare and the available environmental and input variables (rainfall, temperature, soil pH, fertilizer usage, pest incidents, and area harvested). The results show that none of these factors have a meaningful linear relationship with yield in this dataset — all correlation values are close to zero (below 0.01 in magnitude). This indicates that, within the ranges captured here, yield variation is likely driven by factors outside the dataset (such as seed variety, farming practice, or localized conditions) rather than by rainfall, temperature, soil pH, fertilizer quantity, or pest pressure alone.

On pest activity specifically, Soybean and Rice recorded the highest average pest incidents (9.67 and 9.61 respectively), while Wheat recorded the lowest (9.27) — though the overall spread across crops is narrow and consistent with the weak correlation to yield noted above.

6. Dashboard Overview

The accompanying Power BI file (Agriculture_Production_insight.pbix) presents these findings as an interactive dashboard. Key design elements include:

- KPI cards summarising total production, average yield, total area harvested, and fertilizer usage.
- Region and Crop slicers allowing dynamic filtering of all visuals.
- A production trend visual to track output across the 2020–2023 period.
- Comparative bar charts for production and yield across crops and regions.
- A rainfall / temperature / soil pH view for contextual environmental analysis.

Note: dashboard screenshots are best captured directly from Power BI Desktop (File → Export → Export as Image, or the Snipping Tool) and can be inserted into this section to give reviewers a direct visual of the published report. This document is structured so those screenshots can be dropped in under this heading without disturbing the rest of the layout.

7. Recommendations

- Investigate agronomic factors beyond this dataset (seed variety, irrigation method, farming practice) to explain yield variation, since rainfall, temperature, soil pH, fertilizer, and pest incidents show minimal correlation with yield here.
- Review Corn cultivation practices, since it has the largest production volume but the lowest per-hectare efficiency — there may be an opportunity to raise yield without expanding area.
- Maintain balanced regional investment, as production is already well distributed across East, West, North, and South.
- Extend the dataset with additional agronomic variables (seed variety, irrigation type, soil nutrients) in future data collection to enable deeper root-cause analysis of yield performance.

8. Conclusion

This project demonstrates an end-to-end Power BI analytics workflow — from raw CSV data to a structured, interactive dashboard — applied to a real-world agriculture production scenario. The analysis shows a stable, well-balanced production base across regions, crops, and years, while highlighting that yield efficiency is not strongly explained by the environmental and input variables captured in this dataset. These insights, along with the interactive dashboard, provide a solid foundation for further exploration and for data-driven decisions in agricultural planning.

— **Neha Bhatt**